

HOJA 01 AUTOMATIZADA

Cálculo matricial paso a paso: Matrices

| <a href="#">← Menú Principal / Teoría</a> |        |        | <a href="#">← Volver a Asistente / Gráfica</a> |        |         |
|-------------------------------------------|--------|--------|------------------------------------------------|--------|---------|
| BARRA                                     | INICIO |        | FIN                                            |        | DX      |
|                                           | X      | Y      | X                                              | Y      |         |
| 1                                         | 300.00 | 500.00 | 700.00                                         | 0.00   | 400.00  |
| 2                                         | 700.00 | 0.00   | 0.00                                           | 0.00   | -700.00 |
| 3                                         | 0.00   | 0.00   | 300.00                                         | 500.00 | 300.00  |

| BARRA | INICIO |   | FIN |   |
|-------|--------|---|-----|---|
|       | X      | Y | X   | Y |
| 1     | 1      | 1 | 2   | 2 |
| 2     | 2      | 2 | 3   | 3 |
| 3     | 3      | 3 | 1   | 1 |

k1=

| U1X        | U1Y        | U2X        | U2Y        |     |
|------------|------------|------------|------------|-----|
| 15,358.36  | -19,197.95 | -15,358.36 | 19,197.95  | U1X |
| -19,197.95 | 23,997.43  | 19,197.95  | -23,997.43 | U1Y |
| -15,358.36 | 19,197.95  | 15,358.36  | -19,197.95 | U2X |
| 19,197.95  | -23,997.43 | -19,197.95 | 23,997.43  | U2Y |

K1=

| U1X        | U1Y        | U2X        | U2Y        | U3X  | U3Y  |
|------------|------------|------------|------------|------|------|
| 15,358.36  | -19,197.95 | -15,358.36 | 19,197.95  | 0.00 | 0.00 |
| -19,197.95 | 23,997.43  | 19,197.95  | -23,997.43 | 0.00 | 0.00 |
| -15,358.36 | 19,197.95  | 15,358.36  | -19,197.95 | 0.00 | 0.00 |
| 19,197.95  | -23,997.43 | -19,197.95 | 23,997.43  | 0.00 | 0.00 |
| 0.00       | 0.00       | 0.00       | 0.00       | 0.00 | 0.00 |
| 0.00       | 0.00       | 0.00       | 0.00       | 0.00 | 0.00 |

k2=

| U2X        | U2Y  | U3X        | U3Y  |     |
|------------|------|------------|------|-----|
| 36,000.00  | 0.00 | -36,000.00 | 0.00 | U2X |
| 0.00       | 0.00 | 0.00       | 0.00 | U2Y |
| -36,000.00 | 0.00 | 36,000.00  | 0.00 | U3X |
| 0.00       | 0.00 | 0.00       | 0.00 | U3Y |

K2=

| U1X  | U1Y  | U2X        | U2Y  | U3X        | U3Y  |
|------|------|------------|------|------------|------|
| 0.00 | 0.00 | 0.00       | 0.00 | 0.00       | 0.00 |
| 0.00 | 0.00 | 0.00       | 0.00 | 0.00       | 0.00 |
| 0.00 | 0.00 | 36,000.00  | 0.00 | -36,000.00 | 0.00 |
| 0.00 | 0.00 | 0.00       | 0.00 | 0.00       | 0.00 |
| 0.00 | 0.00 | -36,000.00 | 0.00 | 36,000.00  | 0.00 |
| 0.00 | 0.00 | 0.00       | 0.00 | 0.00       | 0.00 |

k3=

| U3X        | U3Y        | U1X        | U1Y        |
|------------|------------|------------|------------|
| 11,439.96  | 19,066.61  | -11,439.96 | -19,066.61 |
| 19,066.61  | 31,777.68  | -19,066.61 | -31,777.68 |
| -11,439.96 | -19,066.61 | 11,439.96  | 19,066.61  |
| -19,066.61 | -31,777.68 | 19,066.61  | 31,777.68  |

K3=

| U1X        | U1Y        | U2X  | U2Y  | U3X        | U3Y        |
|------------|------------|------|------|------------|------------|
| 11,439.96  | 19,066.61  | 0.00 | 0.00 | -11,439.96 | -19,066.61 |
| 19,066.61  | 31,777.68  | 0.00 | 0.00 | -19,066.61 | -31,777.68 |
| 0.00       | 0.00       | 0.00 | 0.00 | 0.00       | 0.00       |
| 0.00       | 0.00       | 0.00 | 0.00 | 0.00       | 0.00       |
| -11,439.96 | -19,066.61 | 0.00 | 0.00 | 11,439.96  | 19,066.61  |
| -19,066.61 | -31,777.68 | 0.00 | 0.00 | 19,066.61  | 31,777.68  |

karmadura

| U1X        | U1Y        | U2X        | U2Y        | U3X        | U3Y        |
|------------|------------|------------|------------|------------|------------|
| 26,798.32  | -131.34    | -15,358.36 | 19,197.95  | -11,439.96 | -19,066.61 |
| -131.34    | 55,775.11  | 19,197.95  | -23,997.43 | -19,066.61 | -31,777.68 |
| -15,358.36 | 19,197.95  | 51,358.36  | -19,197.95 | -36,000.00 | 0.00       |
| 19,197.95  | -23,997.43 | -19,197.95 | 23,997.43  | 0.00       | 0.00       |
| -11,439.96 | -19,066.61 | -36,000.00 | 0.00       | 47,439.96  | 19,066.61  |
| -19,066.61 | -31,777.68 | 0.00       | 0.00       | 19,066.61  | 31,777.68  |

Matriz de cargas:

|     |     |
|-----|-----|
| C1X | U1X |
| C1Y | U1Y |

|           |
|-----------|
| 4,000.00  |
| -5,000.00 |

|           |            |            |          |             |
|-----------|------------|------------|----------|-------------|
| <b>C=</b> | <b>C2X</b> | <b>U2X</b> | <b>=</b> | <b>0.00</b> |
|           | <b>C2Y</b> | <b>U2Y</b> |          | <b>C2Y</b>  |
|           | <b>C3X</b> | <b>U3X</b> |          | <b>C3X</b>  |
|           | <b>C3Y</b> | <b>U3Y</b> |          | <b>C3Y</b>  |

**Matriz de desplazamientos:**

|           |            |            |          |            |   |   |
|-----------|------------|------------|----------|------------|---|---|
| <b>D=</b> | <b>D1X</b> | <b>U1X</b> | <b>=</b> | <b>D1X</b> | 1 |   |
|           | <b>D1Y</b> | <b>U1Y</b> |          | <b>D1Y</b> | 2 |   |
|           | <b>D2X</b> | <b>U2X</b> |          | <b>D2X</b> | 3 |   |
|           | <b>D2Y</b> | <b>U2Y</b> |          | <b>0</b>   |   | 1 |
|           | <b>D3X</b> | <b>U3X</b> |          | <b>0</b>   |   | 2 |
|           | <b>D3Y</b> | <b>U3Y</b> |          | <b>0</b>   |   | 3 |

**Ecuacion matricial:**

$$C = Karmadura * D$$

ELIMINO FILA Y COLUMNA DE LOS QUE NO CORRESPOND

ELIMINO COLUMNA SEGÚN LAS REACCIONES EN ORDEN :

| C         |   | K          |            |            |            |            |
|-----------|---|------------|------------|------------|------------|------------|
| 4,000.00  | = | 26,798.32  | -131.34    | -15,358.36 | 19,197.95  | -11,439.96 |
| -5,000.00 |   | -131.34    | 55,775.11  | 19,197.95  | -23,997.43 | -19,066.61 |
| 0.00      |   | -15,358.36 | 19,197.95  | 51,358.36  | -19,197.95 | -36,000.00 |
| C2Y       |   | 19,197.95  | -23,997.43 | -19,197.95 | 23,997.43  | 0.00       |
| C3X       |   | -11,439.96 | -19,066.61 | -36,000.00 | 0.00       | 47,439.96  |
| C3Y       |   | -19,066.61 | -31,777.68 | 0.00       | 0.00       | 19,066.61  |

**Ecuacion matricial reducida para el calculo de desplazamientos:**

|                  |          |                   |                  |                   |   |            |
|------------------|----------|-------------------|------------------|-------------------|---|------------|
| <b>4,000.00</b>  | <b>=</b> | <b>U1X</b>        | <b>U1Y</b>       | <b>U2X</b>        | · | <b>D1X</b> |
| <b>-5,000.00</b> |          | <b>26,798.32</b>  | <b>-131.34</b>   | <b>-15,358.36</b> |   | <b>D1Y</b> |
| <b>0.00</b>      |          | <b>-131.34</b>    | <b>55,775.11</b> | <b>19,197.95</b>  |   | <b>D2X</b> |
|                  |          | <b>-15,358.36</b> | <b>19,197.95</b> | <b>51,358.36</b>  |   |            |

$$D = \text{MINVERSA}(K) * C$$

| D   |   | MIN          |              |              | C         |
|-----|---|--------------|--------------|--------------|-----------|
| D1X | = | 4.63865E-05  | -5.35431E-06 | 1.58730E-05  | 4,000.00  |
| D1Y |   | -5.35431E-06 | 2.11947E-05  | -9.52381E-06 | -5,000.00 |
| D2X |   | 1.58730E-05  | -9.52381E-06 | 2.77778E-05  | 0.00      |

$$D = \text{MMULT}(\text{MIN}) * C$$

**D**

**MM\*C**

|     |   |              |    |
|-----|---|--------------|----|
| D1X | = | 0.212317513  | cm |
| D1Y |   | -0.127390508 | cm |
| D2X |   | 0.111111111  | cm |

Ecuacion matricial reducida para el calculo de reacciones

| C   |   |            | D          |            |              |
|-----|---|------------|------------|------------|--------------|
| C2Y | = | 19,197.95  | -23,997.43 | -19,197.95 | 0.212317513  |
| C3X |   | -11,439.96 | -19,066.61 | -36,000.00 | -0.127390508 |
| C3Y |   | -19,066.61 | -31,777.68 | 0.00       | 0.111111111  |

|     |   |           |     |
|-----|---|-----------|-----|
| C2Y | = | 5,000.00  | Kgf |
| C3X |   | -4,000.00 | Kgf |
| C3Y |   | 0.00      | Kgf |

Matriz de desplazamientos total

|     |     |              |
|-----|-----|--------------|
| D1X | U1X | 0.212317513  |
| D1Y | U1Y | -0.127390508 |
| D2X | U2X | 0.111111111  |
| D2Y | U2Y | 0.000000000  |
| D3X | U3X | 0.000000000  |
| D3Y | U3Y | 0.000000000  |

Calculo de las fuerzas

|           |            |                        |        |        |         |             |
|-----------|------------|------------------------|--------|--------|---------|-------------|
| AE/L      |            | cx                     | cy     | cx     | cy      | D           |
|           |            | -                      | -      | +      | +       |             |
| 39,355.79 |            | -0.6247                | 0.7809 | 0.6247 | -0.7809 | 0.21231751  |
| F1=       | -6,403.124 |                        |        |        |         | -0.12739051 |
|           |            |                        |        |        |         | 0.11111111  |
|           |            |                        |        |        |         | 0.00000000  |
|           | Kgf        | BARRA A COMPRESIÓN (-) |        |        |         |             |

|      |    |    |    |    |   |
|------|----|----|----|----|---|
| AE/L | cx | cy | cx | cy | D |
|------|----|----|----|----|---|

|           |                      |        |        |         |        |            |
|-----------|----------------------|--------|--------|---------|--------|------------|
|           |                      | -      | -      | +       | +      |            |
| 36,000.00 |                      | 1.0000 | 0.0000 | -1.0000 | 0.0000 | 0.11111111 |
| F2=       | 4,000.000            |        |        |         |        | 0.00000000 |
|           |                      |        |        |         |        | 0.00000000 |
|           |                      |        |        |         |        | 0.00000000 |
|           |                      |        |        |         |        |            |
| Kgf       | BARRA A TRACCIÓN (+) |        |        |         |        |            |

| AE/L      |                      | cx      | cy      | cx     | cy     | D           |
|-----------|----------------------|---------|---------|--------|--------|-------------|
|           |                      | -       | -       | +      | +      |             |
| 43,217.64 |                      | -0.5145 | -0.8575 | 0.5145 | 0.8575 | 0.00000000  |
| F3=       | 0.000                |         |         |        |        | 0.00000000  |
|           |                      |         |         |        |        | 0.21231751  |
|           |                      |         |         |        |        | -0.12739051 |
| Kgf       | BARRA DE FUERZA NULA |         |         |        |        |             |

| RESUMEN GENERAL DE FUERZAS EN LAS BARRAS |          |          |                 |              |                |
|------------------------------------------|----------|----------|-----------------|--------------|----------------|
| BARRA #                                  | NUDO INI | NUDO FIN | LONGITUD L (cm) | ÁREA A (cm²) | FUERZA AXIAL N |
| 1                                        | 1        | 2        | 640.31          | 12.00        | -6,403.124     |
| 2                                        | 2        | 3        | 700.00          | 12.00        | 4,000.000      |
| 3                                        | 3        | 1        | 583.10          | 12.00        | 0.000          |

A - EJEMPLO 1 RESUELTO (WHATSAPP)

s locales, Ensamblaje, Desplazamientos, Reacciones y Fuerzas

| DY      | L      | Cx  | Cy      | A     | E            |
|---------|--------|-----|---------|-------|--------------|
| -500.00 | 640.31 | ### | -0.7809 | 12.00 | 2,100,000.00 |
| 0.00    | 700.00 | ### | 0.0000  | 12.00 | 2,100,000.00 |
| 500.00  | 583.10 | ### | 0.8575  | 12.00 | 2,100,000.00 |

|     |
|-----|
| U1X |
| U1Y |
| U2X |
| U2Y |
| U3X |
| U3Y |

|     |
|-----|
| U1X |
| U1Y |
| U2X |
| U2Y |
| U3X |
| U3Y |

|     |
|-----|
| U1X |
| U1Y |
| U2X |
| U2Y |
| U3X |
| U3Y |

|     |
|-----|
| U1X |
| U1Y |
| U2X |
| U2Y |
| U3X |
| U3Y |

| 1. MATRIZ DE CARGAS Y DESPLAZAMIENTOS ({C} Y {D})             |
|---------------------------------------------------------------|
| • Columna B: Variable del vector (C1X, C1Y / D1X, D1Y).       |
| • Columna C: Grado de libertad global asociado (U1X, U1Y...). |



|     |
|-----|
| U2Y |
| U3X |
| U3Y |

|  |
|--|
|  |
|  |
|  |

|                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------|
| <b>4. CÁLCULO DE REACCIONES: <math>\{R\} = [K_{21}] \cdot \{D\}</math></b>                                                |
| • Submatriz $[K_{21}]$ : Filas correspondientes a grados con apoyo ( $D=0$ ) y columnas correspondientes a grados libres. |
| • Multiplicación [MMULT]: $\{R\} = [K_{21}] \cdot \{D_{libres}\}$ genera las reacciones en cada grado con apoyo.          |
| • Nomenclatura del Profesor: C1X, C1Y, C2Y... representan las fuerzas reactivas en cada grado con apoyo.                  |
| • Verificación de Equilibrio Global: La suma algebraica de reacciones más cargas debe ser cero.                           |
| $\sum F_x = 0$ y $\sum F_y = 0$ (Equilibrio estático exacto).                                                             |
| • Color distintivo: Celdas de reacción identificadas con fondo lavanda pastel (hex: #E6E6FA).                             |
|                                                                                                                           |
|                                                                                                                           |

|     |
|-----|
| D1X |
| D1Y |
| D2X |
| D2Y |

|                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>5. CÁLCULO DE FUERZAS EN BARRAS: <math>\{F_m\} = (AE/L) \cdot [-c_x, -c_y, c_x, c_y]</math></b>                                            |
| • Fila de Signos $[-c_x, -c_y, +c_x, +c_y]$ : Derivada de la deformación axial relativa.                                                      |
| $\Delta L = (u_{fin} - u_{ini}) \cdot c_x + (v_{fin} - v_{ini}) \cdot c_y = -c_x \cdot u_i - c_y \cdot v_i + c_x \cdot u_j + c_y \cdot v_j$ . |
| • Rigidez axial de barra: $AE/L$ (Kgf/cm), donde A = área (cm²), E = módulo elástico (Kgf/cm²).                                               |
| • Fuerza Axial $F_m$ : Obtenida mediante $=(MMULT(C:F, G:G)*A)$ multiplicando la fila de signos por la rigidez.                               |
| • Diagnóstico de Estado Estructural:                                                                                                          |
| - $F_m > +0.001$ Kgf --> BARRA A TRACCIÓN (+) (Esfuerzo de tracción, color rojo).                                                             |
| - $F_m < -0.001$ Kgf --> BARRA A COMPRESIÓN (-) (Esfuerzo de compresión, color verde).                                                        |
| - $ F_m  \leq 0.001$ Kgf --> BARRA DE FUERZA NULA (Barra de estabilidad o sin fuerza, color gris).                                            |
| • Resumen General: Tabla final unificada de 7 columnas que consolida todas las reacciones y fuerzas.                                          |
|                                                                                                                                               |

|     |
|-----|
| D2X |
| D2Y |
| D3X |
| D3Y |

|     |
|-----|
| D3X |
| D3Y |
| D1X |
| D1Y |

| <b>ESTADO ESTRUCTURAL BARRA A COMPRESIÓN (-) TRACCIÓN (+)</b> |
|---------------------------------------------------------------|
| BARRA DE FUERZA NULA                                          |

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There is a vertical margin line on the left side, creating a narrow left margin. The paper appears to be a standard notebook page.



| AE/L      |
|-----------|
| 39,355.79 |
| 36,000.00 |
| 43,217.64 |



...á libre ( $R_x=0$ ),  $C=P_x$  (carga externa de GRAFICA).

$D = D_{nx}$  (desplazamiento incógnita).

cos (D=0) para extraer K11 y K21 sin macros.

quier cambio de apoyo o carga en tiempo real.

$$[global] \cdot \{D\}.$$

PONDE PARA HALLAR LOS DESPLAZAMIENTOS'.

os restringidos (índice 2):

$$\text{alcul ar } \{D\_libres\} = [K11]^{-1} \cdot \{C\_libres\}.$$

5 reacciones  $\{R\} = [K21] \cdot \{D\_libres\}$ .

dura usando los grados de libertad libres.

das dinamicamente de GRAFICA).

(cm/Kgf).

los nodales reales en centímetros (cm).

o hacia arriba; Negativo (-) hacia la izquierda o abajo.

capturar micro-deformaciones elásticas.

|  |
|--|
|  |
|  |
|  |

|                                            |
|--------------------------------------------|
|                                            |
| nas de grados libres (D≠0).                |
| da apoyo restringido en Kgf.               |
| as en los apoyos articulados o de rodillo. |
| gas externas debe ser idénticamente cero:  |
|                                            |
| (#EDEDFA).                                 |
|                                            |
|                                            |

|                                                   |
|---------------------------------------------------|
| <b>c, cy] · {D}</b>                               |
| de la barra m:                                    |
|                                                   |
| istico (Kgf/cm²), L = longitud (cm).              |
| el vector de cosenos por los desplazamientos {D}. |
|                                                   |
| or menta pastel).                                 |
| n, color coral pastel).                           |
| i carga transmitida).                             |
| las barras con sus fuerzas y estados.             |
|                                                   |

